Aurelio Spadotto

0614086871 • aurelio-edoardo.spadotto@umontpellier.fr
aurelio-spadotto.github.io

Research Topics

Numerical analysis for partial differential equations ; Flow-Structure Interactions in biological flows ; Capsule Dynamics ; Electrophisiology of Red Blood Cells ; Polytopal numerical methods applied to internal boundary problems.

Publications

Article: D.A. Di Pietro, S. Mendez, A. Spadotto,
"A discrete de Rham discretization of interface diffusion problems with application to the Leaky Dielectric Model"
Journal of Computional Physics, 2025. Published online, DOI:10.1016/j.jcp.2025.113920

Conferences

Conference: A.Spadotto, S.Mendez, "Immersed boundary simulations of red blood cells in blood analyzers : representation of Maxwell stress", *Journées Math Bio Santé 2024*, Nantes, 2024

Workshop: A.Spadotto, S.Mendez, P.Benard, "Electrodeformation of red blood cells, extension to 3D and improved accuracy at membrane", *Extreme-CFD Workshop, 7-th Ed.*, Merville-Franceville, 2024

Workshop: A.Spadotto, S.Mendez, "Ghost Fluid Method (GFM) for Electrodeformation", *Extreme-CFD Workshop, 6-th Ed.*, Merville-Franceville, 2023

Conference: A.Spadotto, S.Mendez, "Towards modeling of the electro-deformation for red blood cells flowing in Coulter counters", *Dynacaps 2023*, Compiègne, 2023

Software Development Experience

YALESBIO (contributor): High performance, parallel Finite Volume solver for biomedical CFD applications written in **Fortran** (detailed presentation and credits [here]). Development of the solver RBC for the singular dynamics of blood cells *in vitro* and *in silico*. Implementation of a module for the estimation of membrane force induced by electric fields present in blood analysis systems. Implementation of a Ghost Fluid Method for the treatment of immersed boundaries and coupling with an Immersed Boundary Method for the membane-flow interaction.

POLY-MEMB (author): A library for the simulation of electrically-induced membrane dynamics based on polygonal methods and written in **Python**. Implementation of a Hybrid High Order scheme for Stokes equations with interfaces. Implementation of a Discrete De Rham scheme for an elliptic problem with interface (documentation here).

HARDCORE (contributor): A **C++** library for the implementation of hybrid numerical methods such as the Hybrid High Order method (HHO), and the Discrete De Rham method (DDR)

(presentation [here]). Development of a HHO solver for a problem of magnetostatics.

Education

IMAG, Université de Montpellier **Montpellier, France**
Ph.D. in Applied Mathematics *2021-2025*
Thesis : "Numerical Simulation of Electrodeformation of Red Blood Cells"
Supervisors : Simon Mendez, Daniele Di Pietro
Jury president : Vincent Moureau
Reviewers : Lisl Weynans, Alexei Lozinski

IMAG, Université de Montpellier **Montpellier, France**
Internship *2021*
Project : "A hybrid high order method for magnetostatics"
Supervisors : Daniele Di Pietro, Simon Lemaire
Final essay [here]

Politecnico di Milano **Milan, Italy**
Master of Science in Mathematical Engineering *2019–2021*
Focus : Numerical Analysis of PDEs and Scientific Computing

Politecnico di Milano **Milan, Italy**
Bachelor of Science in "Ingegneria Matematica" *2016–2019*

Teaching Experience

Université de Montpellier, Faculté de Science **Montpellier, France**
Teaching Assistant *2022-*

- Raisonnement Scientifique ; spring 2025 (Path : Biology)
- Outils Mathématiques 2 ; fall 2024 (Path : Physics, Chemistry, Engineering Science)
- Outils Mathématiques 1 ; fall 2024, fall 2023 (Path : Physics, Chemistry, Engineering Science)
- Méthodes Calculatories ; fall 2023, fall 2022 (Path : Biology)

Complementary Skills

Software Tools: GIT for version control ; Paraview for 3D visualization.

References

Simon Mendez: (Ph.D. supervisor, IMAG) : simon.mendez@umontpellier.fr

Daniele Di Pietro: (Ph.D. supervisor, IMAG) : daniele.di-pietro@umontpellier.fr